Lesson 9: Project: Secret Messenger

Module 3 | 50 minutes | Project: cryptography app | Grades 6–8 · ages 12–14

Name: _____ Class: _____ Date: _____

Plan your project and practise the key ideas.

1. Key words

Match each word to what it means. Write the letter.

a. encrypt / decrypt ___ A. Encrypt then decrypt to check you get the original.
b. inverse ___ B. An operation that undoes another.
c. round-trip test ___ C. Scramble / unscramble a message.

2. Round trip

Order the round-trip test Number them 1–5.

a. ___ Decrypt it with the same key
b. ___ Copy the encrypted text
c. ___ Encrypt the message
d. ___ Choose a message and a key
e. ___ Check you got the original message

3. My plan

Plan your project: Build a messenger that encrypts and decrypts with a Caesar cipher: a menu, message and key inputs, one shift procedure used for both directions, validation, and round-trip tests. Sketch it and list what each part will do.

4. Checklist

Tick each one when your project does it.

a. ☐ Menu
b. ☐ Message and key inputs
c. ☐ Shift procedure
d. ☐ Wrap-around
e. ☐ Validation
f. ☐ Round-trip tests

5. Two stars and a wish

Ask a friend to test your project. Write two things they liked and one idea to improve it.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ My app encrypts and decrypts with a key.
☐ It uses one shift procedure.
☐ Round-trip tests pass.